

Review: Identifying congruent triangles



1 **A** and **B**

2 **B** and **D**

3

a

i **B** and **C**

ii SSS: All three corresponding sides are congruent.

b

i **A** and **C**

ii SAS: A pair of corresponding sides and the included angle are congruent.

c

i **B** and **C**

ii ASA: A pair of corresponding angles and the included side are congruent.

d

i **A** and **C**

ii SAS: A pair of corresponding sides and the included angle are congruent.

4

a

i **A** and **B**

ii SAS: A pair of corresponding sides and the included angle are congruent.

b

i **B** and **C**

ii AAS: A pair of corresponding angles and a non-included side are congruent.

c

i **C** and **D**

ii HL: Two right-angled triangles with hypotenuse and one leg are congruent.

d

i **C** and **D**

ii SAS: A pair of corresponding sides and the included angle are congruent.

5

a No

b No

c No

d No

6 No



7 No

8

a $\overline{DF}$

b SSS: All three corresponding sides are congruent.

9

a $\angle CDB$

b SAS: A pair of corresponding sides and the included angle are congruent.

10

a $\angle RQS$

b ASA: A pair of corresponding angles and the included side are congruent.

11

a $\overline{PN}$

b SSS: All three corresponding sides are congruent.

12

a $\angle CDB$

b SAS: A pair of corresponding sides and the included angle are congruent.

13

a $\angle POQ$

b AAS: A pair of corresponding angles and a non-included side are congruent.

14

a $\triangle PRN$ and $\triangle QRM$

b $\triangle EAB$ and $\triangle BDE$

15 One triangle could be an enlarged or reduced version of the other.

16 No, not always.



Additional questions

17

- a $\triangle ABC$ and $\triangle EDC$ are congruent. $\triangle EDC$ is a rotation of $\triangle ABC$ by 180° around point C .
- b $\triangle PQR$ and $\triangle SQR$ are congruent. $\triangle SQR$ is a reflection of $\triangle PQR$ across $\overleftrightarrow{QR}$.
- c $\triangle UVW$ and $\triangle XYZ$ are not congruent.

18

- a Congruent by Side-angle-side congruence (SAS)
- b Congruent by Side-side-side congruence (SSS)
- c Not congruent
- d Not enough information
- e Congruent by Angle-angle-side congruence (AAS)
- f Not congruent
- g Not enough information
- h Not congruent
- i $\triangle ABC \cong \triangle DEF$ based on the HL congruence theorem.
- j There is not enough information to determine whether $\triangle GHJ \cong \triangle KLM$ are congruent.
- k $\triangle ABC$ and $\triangle STU$ are not congruent.
- l $\triangle PQR$ and $\triangle STU$ are congruent based on the Side-angle-side (SAS) congruence theorem.

19

- | | |
|-------------------------|------------------|
| a $x = 11.5$ | b $x = 90^\circ$ |
| c $x = 7$ | d $x = 3$ |
| e $x = 9$ | f $x = 4$ |
| g $x = 26$ | h $x = 60^\circ$ |
| i $x = 19$ and $y = 18$ | |

20

- a There is enough information for Side-angle-side (SAS) congruence
- b Answers may vary.
There is not enough information for congruence. If $\angle E \cong \angle B$ the triangles would be congruent by Side-angle-side congruence (SAS)

- c There is not enough information for congruence. If $\overline{EG} \cong \overline{HK}$ the triangles would be congruent by Side-side-side congruence (SSS) or if $\angle F \cong \angle J$ the triangles would be congruent by Side-angle-side congruence (SAS).
- d There is enough information for Side-side-side congruence (SSS).

21

- a
- i $\triangle ACD$ and $\triangle BCE$
 - ii Side-side-side congruence (SSS)
- b Answers may vary:
- i $\triangle PRN$ and $\triangle QRM$
 - ii Side-side-side congruence (SSS) or Side-angle-side congruence (SAS)
- c Answers may vary:
- i $\triangle VZY \cong \triangle YXV$
 - ii Hypotenuse-leg (HL)

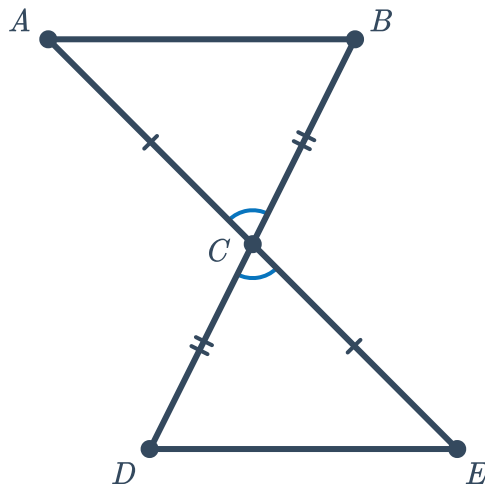
22

- a
- i $\angle C \cong \angle B$
 - ii $\angle BAD \cong \angle CAD$
- b
- i Either $\overline{AC} \cong \overline{JM}$ or $\overline{AB} \cong \overline{KM}$
 - ii $\overline{BC} \cong \overline{JK}$

23

- a $\overline{UV} \cong \overline{WX}$
- b $\angle CED \cong \angle FDE$

- 24 Yes. The intersecting lines form a pair of congruent vertical angles. Because the segments bisect each other, the triangles will be congruent by Side-angle-side congruence (SAS) no matter how long the segments are.





- 25** No. While it is true that the longest side of a right-triangle is the hypotenuse, the triangles that Fleur and Jeung have drawn are not necessarily right-triangles. Just knowing that two pairs of sides are congruent is not enough to prove congruency.
- 26** Yes. If the side is in between the two given angles students will be able to construct a congruent triangle by Angle-side-angle congruence (ASA) and if the side is *not* between the two angles the triangles will be congruent by Angle-angle-side congruence (AAS).